Sello Ramdeps - Exercise 4

SQL Practice Exercise Joins, union, filtering and aggregate

Question 1 .

Retrieve all employees and their assigned projects using an Inne JOIN

SQL

Select employee ID, first-name, lastname, Department, Salary, Project id
Projectname, budget, status

From Employees

Inner Joins projects P on e.EmployeeID = P Employee ID

Outcome

ID	First name	Last name	Dept	Salary	Project ID	Project name	Budget	Status
1	John	Doe	IT	70 000	101	IDevelopment	100 000	Completed
1	John	Doe	IT	70 000	103	Cybersecurity	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80000	106	Risk	80 000	Pending

Question 2

② Retrieve all employees and their assigned projects including Employees who have no projects using ~~left~~ Left ~~Right~~ join

SQL

```
Select Employee ID, ~~project~~ First name, lastname, Departments, Salary
Project ID, ProjectName, budget, status

From Employees

Left Join project P ON E.Employee = P.Employee ID
```

Em ID	First Name	Last Name	Dept	Salary	Project ID	Project name	Budget	status
1	John	Doe	IT	70 000	101	A1	108 000	Completed -
1	John	Doe	IT	70000	103	Cyber	75 000	Pending
2	Smith	Smith	HR	60000	102	Employee Training	50 000	Ongoing
3	Johnson	Finance	Finance	75000	104	Financial Analyses	90 000	Null
4	Brown	IT	IT	72000	Null	Null	NULL	Completed
5	Wilson	Sales	Sales	65 000	105	Market	65000	Completed Pending
6	Clark	Finance	Finance	80000	100	Risk	80000	Pending

Question 3

Retrieve all projects and their assigned employees, including projects that have no Employees using a right join

SQL

```
select, ProjectID, Projectname, budget, status, EmployeeID, First name
lastName, Department, Salary
From Employees
RightJoin project P On e.employeeID = P.Employee ID
```

Project ID	Project name	Budget	Status	Employee ID	first name	Last name	Department	Salary
101	AI Deve		Completed	1	John	Doe	IT	70 000
102	Employee Training		Ongoing	2	Alice	Smith	HR	60 000
103	Cyber Security		Pending	1	John	Doe	IT	70 000
104	Financial Analysis		ongoing	3	Bob	Johnson	Finance	75 000
105	Market Expansion		Completed	5	Emmy	wilson	Sales	65 000
106	Risk		Pending	6	Mike	clark	Fire	80 000

Budget
100 000
50 000
75 000
40 000
65 000
80 000

Question 4

SQL

```
select EmployeeID, Firstname, lastname, Department, Salary, project ID
Projectnames, budget, status
From Employees
FullOuterJoin Project P on EmployeeID = P Employee ID
```

The result will be the same as left Join

(2) UNION & UNION ALL

(5) Retrieve a list of all unique cities where employees are located and project statues

* Return: Location (Rename the Column to location using an Alias)

SQL

```
Select city as location from Employees
union
select Status AS Location from project
```

Location

New York
Los Angeles
Toronto
London
Sydney
Completed
ongoing
Pending

Question 6

Retrieve a list of all cities where employees are located and project statuses allowing duplicates

SQL

```
Select city AS location from employees
union ALL
Select status AS location from projects;
```

Location	
Newyork	Completed
Los Angeles	Ongoin
Toronto	Pending
London	ongoing
sydney	completed
Newyork	pend

Question 7

Filtering statements

7. Retrieve employees who earn more than 70000
Return: Employee ID, firstname, lastname, Department, Salary

SQL

```
Select, EmployeeID, Firstname, lastname, Departments, salary
from Employees
where Salary > 70000
```

Employee ID	First name	Last name	Department	Salary
3	Bob	John	Finance	75000
4	David	Brown	IT	72000
5	Michel	Clark	Finance	80000

Question 8
Filtering statement

8) Retrieve employees working in either IT or finance Department

SQL
```
select, EmployeeID, firstname, Lastname, Department, Salary
from Employees
where depart between IT and finance >
```

Employee ID	Firstname	Lastname	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michell	Clark	Finance	80 000

Question 9

Retrieve projects that are not yet complete

SQL
```
Select, ProjectID, Projectname, budget, status
from project
where status = completed
```

Project ID	Project name	Budget	Status
102	Employee Training	50 000	ongoing
103	Cyber Security	75 000	pending
104	Financial Analysis	40 000	ongoing
106	Risk Management	80 000	pend n

Question 10

1 Retrieve projects that have a budget greater than 70 000 and are not completed

SQL select Projectid, Projectname, Budget, Status

from project

Where project > 70 000 and Status = Completed

Project ID	Project name	Budget	Status
103	Cyber Security	75 000	Pending
104	Financal Analys	90 000	ongoing
106	Risk management	80 000	Pending

Question 11

Retrieve Employees from newyork or Toronto ordered by Salary in descending order

SQL select EmployeeID, first name, lastname, Departme, Salary City

from Employee

order by desc

Limit 3

Employee ID	First name	Lastname	Department	Salary	City
6	Mandeep	clark	Finance	8000	NY
3	Bob	John	Finance	75 000	Toronto
1	John	Doe	IT	70000	NY

Question 12

Retrieve the top 3 highest - Paid employees
Employee ID, Firstname, lastname, Department, Salary

from employee
order by desc;
limit 3

Employee ID	First name	last name	Department	Salary
3	Bob	Doe	IT	75 000
4	David	brown	IT	72 000
6	Michel	Clark	Finance	60 000

Question 13

Aggregate functions with Group by & having

3) Find the total salary per department sorted in descending order

Sol Select Department, Sum(Salary) As Total salary
From Employees
Group by department
Order by total salary desc;

Department	Total Salary
Finance	155 000
IT	142 000
Sales	65 000
HR	60 000

Question 14

Find the average salary per city, but only include cities where average salary is greater than 65 000

SQL

```
Select, Department, Sum(Salary) As Total Salary
From Employees
Group by Department
order by totalSalary Desc
```

Department	Total salary
Finance	155 000
IT	142 000
Sales	65 000
HR	60000

Question 15

Count the number of employees per department including only departments with more than 1 employee

SQL

```
Select department, Count(employeeID) As Employee Count
From Employees
Group by department
Having Count(employeeID) > 1
```

Department	Employee Count
IT	2
Finance	2

Question 16

Retrieve the number of projects per status, but only include ~~StateSes~~ with atleast 2 project

SQL

```
select, count (projectID) as project count
from projects
Group by status
Having Count (projectID) >=2
```

status	project count
Completed	2
Ongoing	2
Pending	2

Question 17

Retrieve the total project budget per employee but only the employees who are managing projects worth more than 150 000

```
Select EmployeeID, Firstname, lastname, totalprojectbudget
from Employee
join project p on EmployeeID = p.EmployeeID
Group by employeeID, e.firstname, e.lastname
Having Sum(p.budget) > 150 000
```

Employee ID	firstname	lastname	Total project budget
1	John	Doe	175000